



Activity File: Creating and Restoring Backups with `tar`

Part 1: Creating Backups with `tar`

In this two-part activity, you are a junior administrator at **Rezifp Pharma Inc.**

- The company conducts clinical trials on drugs for oncology, immunology, and vaccines. In recent weeks, there have been a series of malware attacks. The company is now strengthening its backup activities.
- In response to the malware attacks, your department has decided to create daily **full backups** of the files associated with the E-Prescription Treatment database, which is the main system for many departments.

You have been tasked with:

- Creating a name for the `tar` archive using the department's standard naming convention.
- Creating daily full backups of the directories and files in the `~/Documents/epscript` directory.
- Printing the file permission, owner, size, date, and time for each file in the archive.
- Verifying the archive after it is written to check for errors.
- Creating a file containing the output of the `tar` command for later review by the SysOps team, which will check file structure, permissions, and errors.

Lab Environment

- To complete the activity, log into this Module's lab environment using the following credentials:
 - Username: `sysadmin`
 - Password: `cybersecurity`

Instructions

1. Move to the `~/Documents/epscript` directory.
2. List the `epscript` directory contents and answer the following question:
 - What directories and files are located there?
3. Standardize the file name for your backup:

- Your department uses the [ISO 8601](#) standard for representing the date in the naming convention for all archives.
 - Using the standard for representing the date, YYYY-MM-DD, allows sysadmins to locate an archive quickly.
 - Use the date **May 5, 2019** and convert it to the ISO 8601 format **without dashes**.
 - Add the filename `epscript.tar` to the end of the ISO 8601 date.
 - What will be the archive name?
4. Write a `tar` command that creates an archive with the following characteristics:
- Recall that `[ISO_8601_date]epscript.tar` is the archive file name.
 - The file backup will include the `epscript` directory and includes all directories and files contained within it.
 - File permission, owner, size, and date and time information are recorded for each file in the archive.
 - The archive is **verified** after writing it, in order to validate the integrity of the data.
Hint: This is a new option. [See the man page](#).
 - The output from the `tar` command is written to the file `[archive_name].txt` for later review by the SysOps team to check file structure, permissions, and errors.
 - Note: `archive_name` is the `tar` archive name you created using `[ISO_8601_date]epscript.tar`.
5. Using the `less` command, review the output of the `.txt` file.
- What is displayed in the output file?

Part 2: Restoring Backups with `tar`

The E-Prescription Treatment database was attacked and taken offline.

- Fortunately, the team had a recent full backup and was able to recover the database, making the system operational. However, a pharmacy technician noticed that some files in the patient database were missing, and the team discovered that the wrong full backup was used. The system was taken offline again.
- It is critical that the missing patient files are restored as soon as possible. You have spoken to the pharmacy technician and received a list of the names of patients whose files are missing.

You have been tasked with:

- Listing the contents of an archive to determine what it contains.
- Creating a directory to restore the patient data for review.
- Extracting only the patient files to the directory so they can be checked by the pharmacy technician.

Instructions

1. Move to the `~/Documents/epscript/backup` directory.

2. List the contents of the directory to display the `20190814epscript.tar` file. You will search this archive for the missing patient data.
3. Extract the patient directory from the `20190814epscript.tar` archive.
 - Extract patient data to the `patient_search` directory in the `~/Documents/epscript/backup` directory.
 - Test for any errors when extracting. This will check the integrity of the archive.
4. List the contents of the `patient_search` directory to check that the patient directory and files were extracted there.

Bonus

5. View the specific patient directory and file information contained within the archive.
 - Use `grep` to find the following two patients' file information located in the archive:
 - Mark Lopez
 - Megan Patel
-